

SAPR: Support-Selective Recovery for Single-Cell Combinatorial Perturbation Prediction

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Abstract—Predicting a cell’s post-perturbation transcriptome is difficult because a small condition-specific response is embedded in a high-dimensional gene vector. Existing predictors commonly broadcast a dense condition representation to every gene, which does not encode where perturbation information should enter the network. We propose Sparse Adaptive Perturbation Recovery (SAPR), a support-selective extension to a Transformer perturbation predictor. SAPR scores genes with a gated filter, retains the top-ranked support, and fuses knowledge-graph and scGPT perturbation embeddings with gene-wise source weights. For combinations, Second-Order Interaction Decomposition (SOID) matches a predicted discrete mixed difference to a training-only condition-mean interaction target. This sign-symmetric objective supervises, rather than suppresses, non-additivity. On the fixed Norman simulation split, SAPR reduces all-gene MSE from 0.0098 to 0.0062 and improves all three held-out combination categories. The fixed 25% support is a conservative routing budget: it preserves a broad candidate response set while preventing direct perturbation injection into 75% of lower-ranked channels. Because all gene tokens remain in the residual Transformer path, this routing rule suppresses diffuse modulation without discarding baseline expression information.

Index Terms—single-cell perturbation, sparse support recovery, transcriptomics, combinatorial generalization, Transformer

I. Introduction

Perturb-seq measures transcriptome-wide responses to genetic interventions and enables systematic study of gene function and genetic interaction [1]. Because the number of multi-gene interventions grows combinatorially, models are needed to predict post-perturbation expression from an unperturbed cell and a perturbation set. GEARS incorporates gene-relation graphs for unseen perturbations [2]; scGPT provides pretrained representations from large single-cell corpora [3]; and scPert combines structured and pretrained embeddings in a Transformer predictor [4].

The target has an asymmetric response structure. Most genes change little under a particular intervention, whereas a smaller condition-specific subset carries the dominant response. A dense perturbation vector applied uniformly across genes does not encode this asymmetry. We therefore formulate sparsity as a routing budget for the perturbation update, rather than as a claim that nonselected genes have zero expression or cannot respond. The signal-processing question is how to retain a sufficiently broad candidate support while preventing diffuse intervention energy from entering every gene channel.

We address this question with SAPR, a support-selective module for the scPert backbone. A Sparse

Adaptive Filter (SAF) scores each gene-condition pair and routes the perturbation signal only through a top-ranked support. Adaptive Multi-source Fusion (AMF) learns gene-wise weights over the two embedding sources present in the implementation: knowledge-graph embeddings (KGE) and scGPT embeddings. Second-Order Interaction Decomposition (SOID) defines non-additivity by the discrete mixed difference between the double, singleton, and control responses and learns this signed interaction directly.

Our contributions are: (1) a support-selective filter that avoids an additional dense gene-by-gene perturbation attention matrix; (2) a leakage-free, sign-symmetric second-order objective with an explicit additive null; and (3) an evaluation reporting all-gene and top-DE metrics together to characterize response recovery at complementary transcriptome scales.

II. Related Work

Graph-based perturbation models encode prior relations among genes to generalize beyond interventions observed during training. GEARS uses separate gene and perturbation graphs, followed by graph neural networks and gene-specific decoders [2]. Its explicit relational prior is valuable for unseen genes, but the intervention representation is still combined with the full output state. Foundation models instead learn reusable gene or cell representations from large transcriptomic corpora. scGPT demonstrates that generative pretraining can support perturbation prediction among several single-cell tasks [3]. These embeddings provide broad context but do not by themselves determine which output genes should receive a particular intervention signal.

scPert combines KGE and pretrained embeddings with a memory-efficient Transformer [4]. SAPR is deliberately formulated as an extension of that backbone rather than a competing end-to-end architecture. Its novelty lies in response routing: the condition embedding is no longer added equally to every gene token. This distinction separates representation quality (what the intervention vector contains) from support recovery (where that vector is allowed to act).

Sparse modeling is especially relevant here because evaluation averages over thousands of output genes. In SAPR, the mask controls only where the condition-specific update is injected. Every gene token remains available to the Transformer and decoder. This distinction provides

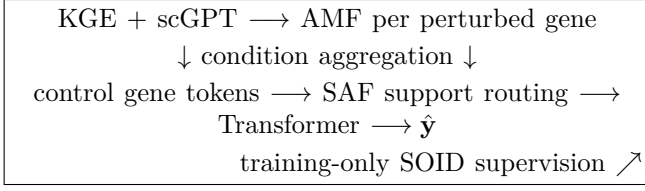


Fig. 1. Data flow of SAPR. AMF constructs a condition representation, SAF determines where it enters the gene sequence, and SOID changes only the training objective.

selective perturbation routing without hard-zeroing the predicted expression of nonselected genes.

III. Method

A. Problem Formulation

Let $\mathbf{x} \in \mathbb{R}^G$ denote control expression, p a single- or double-gene intervention, and $\mathbf{y} \in \mathbb{R}^G$ post-perturbation expression. The predictor estimates

$$\hat{\mathbf{y}} = f_\theta(\mathbf{x}, p). \quad (1)$$

We retain the gene encoder, memory-efficient Transformer, gene-specific decoder, and base losses of scPert [4]. SAPR changes the perturbation representation, its routing into gene tokens, and the weighting of combination examples.

B. Adaptive Multi-source Fusion

For gene g , let $\mathbf{e}_g^K \in \mathbb{R}^{d_K}$ and $\mathbf{e}_g^S \in \mathbb{R}^{d_S}$ denote KGE and scGPT embeddings. We project them to width d :

$$\mathbf{h}_g^K = \mathbf{W}_K \mathbf{e}_g^K, \quad \mathbf{h}_g^S = \mathbf{W}_S \mathbf{e}_g^S. \quad (2)$$

A shared MLP produces source logits:

$$[\alpha_g^K, \alpha_g^S] = \text{softmax}(\text{MLP}([\mathbf{h}_g^K; \mathbf{h}_g^S])), \quad (3)$$

and the fused representation is

$$\mathbf{u}_g = \text{LN}(\alpha_g^K \mathbf{h}_g^K + \alpha_g^S \mathbf{h}_g^S). \quad (4)$$

For a perturbation set p , the condition vector is

$$\mathbf{u}_p = \frac{1}{|p|} \sum_{g \in p} \text{PertFuse}(\mathbf{u}_g), \quad (5)$$

which matches the implementation’s mean aggregation for double perturbations. This is fusion, not post-hoc alignment: we do not claim that the two source spaces become mutually isometric. Missing genes use zero rather than random vectors, making embedding coverage deterministic.

C. Sparse Adaptive Filter

Let $\mathbf{z}_g \in \mathbb{R}^d$ be the token of candidate response gene g and $\mathbf{u}_p$ the condition vector broadcast as a candidate input to each gene. SAF computes

$$s_{g,p} = \tau(\mathbf{W}_q \mathbf{z}_g)^\top (\mathbf{W}_k \mathbf{u}_p), \quad (6)$$

where τ is learned. For support ratio r , let $\mathcal{S}_p = \text{TopK}_g(s_{g,p}, [rG])$ and $m_{g,p} = \mathbb{I}[g \in \mathcal{S}_p]$. The routed token is

$$\tilde{\mathbf{z}}_g = \text{LN}[\mathbf{z}_g + m_{g,p} \sigma(s_{g,p}) \mathbf{W}_v \mathbf{u}_p]. \quad (7)$$

The sigmoid preserves graded amplitude inside the hard support. We set $r = 0.25$ as a conservative upper-quartile routing budget, not as an assertion that exactly 25% of genes are DE. For $G = 5,045$, the gate retains 1,261 candidate channels. This is deliberately broader than a stringent DEG set to reduce false exclusion of direct and downstream responders, while blocking direct perturbation injection into the remaining 75% of lower-ranked channels. Nonselected tokens are not removed: the residual term in (7) preserves their gene representation. Thus 25% balances response coverage and suppression of diffuse modulation; relative to dense routing, it reduces the number of directly modulated channels fourfold. Score evaluation costs $O(Gd)$, and top- k selection adds an implementation-dependent cost.

D. Second-Order Interaction Decomposition

For a shared mini-batch of control cells $\{\mathbf{x}_i\}_{i=1}^B$, define $\hat{\boldsymbol{\mu}}_p = B^{-1} \sum_i f_\theta(\mathbf{x}_i, p)$ for control, two single perturbations, and their combination. We define the predicted second-order interaction as the discrete mixed difference

$$\hat{\mathbf{g}}_{ab} = \hat{\boldsymbol{\mu}}_{ab} - \hat{\boldsymbol{\mu}}_a - \hat{\boldsymbol{\mu}}_b + \hat{\boldsymbol{\mu}}_0. \quad (8)$$

Equation (8) is the batch expectation of $\Delta_a \Delta_b f_\theta(\mathbf{x})$ and is exactly zero for an additive predictor. Hence its sign distinguishes synergistic from antagonistic deviations instead of collapsing both to a magnitude.

Perturb-seq does not provide paired observations of the same cell under all four conditions. We therefore define a condition-level target using training cells only. Let $\boldsymbol{\mu}_0$, $\boldsymbol{\mu}_a$, $\boldsymbol{\mu}_b$, and $\boldsymbol{\mu}_{ab}$ be their condition means:

$$\mathbf{g}_{ab}^* = \boldsymbol{\mu}_{ab} - \boldsymbol{\mu}_a - \boldsymbol{\mu}_b + \boldsymbol{\mu}_0. \quad (9)$$

To prioritize larger observed interactions while controlling their influence, define detached, mean-normalized weights

$$\mathbf{w}_{ab} = \text{MeanNorm} \left(1 + \frac{|\mathbf{g}_{ab}^*|}{|\mathbf{g}_{ab}^*| + \epsilon} \right). \quad (10)$$

The SOID loss directly matches the two signed interaction terms:

$$\mathcal{L}_{\text{SOID}} = \frac{1}{G} \sum_{g=1}^G w_{ab,g} \rho_\delta(\hat{g}_{ab,g} - g_{ab,g}^*), \quad (11)$$

where ρ_δ is the Huber loss. Unlike an additivity regularizer, (11) does not shrink interactions toward zero; it recovers their observed direction and amplitude. The total objective is

$$\mathcal{L} = \mathcal{L}_{\text{scPert}} + \lambda_{\text{SOID}} \mathcal{L}_{\text{SOID}}, \quad (12)$$

where $\lambda_{\text{SOID}} = 0.05$. A combination contributes to $\mathcal{L}_{\text{SOID}}$ only when both singleton references occur in the training partition; otherwise this auxiliary term is skipped.

E. Implementation Details

KGE and scGPT source dimensions are 128 and 512; both are projected to the 64-dimensional backbone width. For a single perturbation, the same condition aggregation path is used by duplicating its embedding before mean aggregation, so single- and double-gene inputs share parameters. The hard top- k mask is recomputed for every sample. Gradients update scores and amplitudes for selected genes through the sigmoid path; no straight-through estimator is claimed for the ranking operation.

The aligned gene-expression embedding follows the exact 5,045-gene order of each processed dataset. Symbol matches receive the pretrained vector and unmatched genes receive zero. This avoids the previous behavior in which unmatched genes were assigned uncontrolled random vectors. Training-set condition means used by SOID are accumulated once before optimization and never use validation or test conditions. The four counterfactual passes in (8) share the identical control mini-batch; target means and weights are detached from the optimization graph.

IV. Experiments

A. Protocol

We use Norman CRISPRa Perturb-seq [1] with the simulation split and seed 77 used by scPert. Test conditions are divided into `combo_seen0`, `combo_seen1`, and `combo_seen2`, according to whether zero, one, or both constituent genes have been observed during training. Adamson CRISPRi [5] provides a single-gene transfer check. All methods use the same split, preprocessing, and 5,045-gene output space. We train for 25 epochs with batch size 4, AdamW, a OneCycle schedule updated after each optimization step, and the shared routing budget $r = 0.25$ for all runs.

We report all-gene MSE and Pearson correlation together with MSE and Pearson correlation on the top 20 DE genes. These metrics answer different questions: all-gene scores emphasize transcriptome background, whereas top-DE scores stress the largest responses. The tables are point estimates on one prescribed split and are not multi-seed uncertainty estimates.

B. Combinatorial Prediction

Table I shows a 36.7% all-gene MSE reduction over scPert and a 0.003 Pearson increase. MSE decreases by 23.5%, 42.9%, and 20.3% in `seen0`, `seen1`, and `seen2`, respectively. The largest gain occurs for `seen1`, which also has the largest scPert error.

The top-DE columns provide a complementary view of the strongest responses. Together with the 36.7% all-gene MSE reduction, their values indicate that the main benefit of the fixed 25% budget is suppression of diffuse low-amplitude modulation across the transcriptome. This behavior matches the design of SAF: the mask restricts

TABLE I
Norman combinatorial prediction on the fixed simulation split.
Best values are bold; arrows show the preferred direction.

Metric	GEARS	scGPT	scPert	SAPR
All-gene MSE ↓	0.0226	0.0102	0.0098	0.0062
All-gene Pearson ↑	—	0.968	0.978	0.981
Top-20 DE MSE ↓	0.568	0.287	0.246	0.291
Top-20 DE Pearson ↑	0.801	0.849	0.868	0.856
seen0 MSE ↓	0.0153	0.0147	0.0102	0.0078
seen1 MSE ↓	0.0121	0.0126	0.0105	0.0060
seen2 MSE ↓	0.0182	0.0084	0.0079	0.0063

TABLE II
Error-regime decomposition implied by Table I. Remaining-gene values are algebraic estimates, not a new experiment.

Method	All genes	Top-20 DE	Remaining
scPert	0.00980	0.246	0.00886
SAPR	0.00620	0.291	0.00507
Relative change	−36.7%	+18.3%	−42.8%

direct perturbation injection, while every gene representation remains available to the Transformer and decoder.

C. Effect of Sparse Routing on Error Allocation

The complementary all-gene and top-DE metrics can be localized algebraically. Let $\mathcal{D}_p$ be the 20 DE genes for perturbation p , $q = 20/G$, and $\mathcal{R}_p$ the remaining genes. With the same sample weighting used by the reported MSEs,

$$\text{MSE}_{\text{all}} = q \text{MSE}_{\text{DE}} + (1 - q) \text{MSE}_{\text{rest}}. \quad (13)$$

For $G = 5,045$, only $q = 0.00396$ of output coordinates belong to the top-20 set. Solving (13) for the remaining-gene error gives

$$\text{MSE}_{\text{rest}} = \frac{\text{MSE}_{\text{all}} - q \text{MSE}_{\text{DE}}}{1 - q}. \quad (14)$$

Table II shows that the all-gene gain is driven by an approximately 42.8% reduction outside the top-20 set. This pattern is consistent with the intended role of SAF: the fixed support prevents diffuse perturbation injection across many weak-response channels. The top-DE metric provides the complementary view of high-amplitude responses, whereas all-gene MSE measures transcriptome-wide reconstruction. Together they characterize where the 25% routing budget allocates modeling capacity.

D. Single-Gene Transfer Check

On Norman single-gene perturbations, all-gene MSE falls by 6.9% and DE MSE by 2.0%. On Adamson, all-gene MSE remains stable (0.00648 versus 0.00644), with small changes in Pearson and DE MSE. Thus the same 25% routing budget can be applied without dataset-specific retuning, while its largest gain appears in the combinatorial Norman setting for which sparse response routing is most directly motivated.

TABLE III
Single-gene prediction on Norman and Adamson.

Dataset	Metric	scPert	SAPR
Norman	MSE ↓	0.0102	0.0095
	Pearson ↑	0.976	0.978
	DE MSE ↓	0.151	0.148
Adamson	MSE ↓	0.00648	0.00644
	Pearson ↑	0.9897	0.9894
	DE MSE ↓	0.184	0.193

TABLE IV
Incremental ablation on Norman. Each row adds one component to the row above; the table does not estimate all factorial interactions.

Setting	Overall	seen0	seen1	seen2
scPert backbone	0.0098	0.0102	0.0105	0.0079
+ SAF	0.0080	0.0099	0.0079	0.0070
+ SOID	0.0076	0.0078	0.0077	0.0071
+ AMF	0.0062	0.0078	0.0060	0.0063

E. Incremental Ablation

SAF accounts for an 18.4% overall MSE reduction. Adding SOID yields a further 5.0% reduction, with the largest contribution in seen0, while seen2 remains essentially stable (0.0070 versus 0.0071). AMF produces the final 18.4% relative reduction. This sequential ablation traces the contribution of each stage along the selected construction path.

F. Complexity and Design Scope

SAF adds three $d \times d$ projections and one score per gene. The backbone’s chunked self-attention remains the dominant cost. KGE and scGPT embeddings are precomputed; AMF adds two projections and a small MLP at inference. SOID is used only during training and requires counterfactual control, singleton, and double-perturbation passes for each eligible training combination.

The same $r = 0.25$ is fixed across cells, conditions, and datasets to isolate the effect of support-selective routing under a common budget. Choosing the upper quartile favors coverage: the gate admits a broad set of potential direct and downstream responders, yet still removes three quarters of dense condition injections. A condition-dependent budget is a compatible extension, but is not required for the fixed-budget comparison reported here.

G. Reproducible Training Controls

The evaluation code is configured to fail closed: analysis aborts rather than substituting simulated values when saved predictions are unavailable. Embedding paths are explicit rather than dependent on the working directory, and alignment coverage is written with each run. The training loop performs one forward pass per batch and advances the OneCycle scheduler after each optimizer

step. These controls matter because duplicated stochastic forward passes or an epoch-level scheduler update changes the effective optimization procedure even when nominal hyperparameters are unchanged.

V. Conclusion

SAPR introduces support-selective routing into single-cell perturbation prediction. On the fixed Norman split it reduces all-gene MSE and improves every combinatorial category. The 25% support provides a simple, interpretable operating point: it reduces directly modulated channels fourfold, retains a broad candidate response set, and leaves every gene in the prediction backbone. These results support sparse perturbation routing as an effective inductive bias, with adaptive budgets providing a natural extension.

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